Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/02/2015
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	Development and Testing of Biomarkers to Determine Individual Astronaut Vulnerabilities to Behavioral Health Disruptions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	08/01/2014	**End Date:**	07/31/2017
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date shows as 7/31/2017 per NSSC information (Ed., 6/2/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barger, Laura Ph.D. (Brigham & Women's Hospital) Czeisler, Charles M.D., Ph.D. (Brigham and Women's Hospital/Harvard Medical Center) Duffy, Jeanne Ph.D. (Brigham & Women's Hospital) Kristal, Bruce Ph.D. (Brigham & Women's Hospital)		
Grant/Contract No.:	NNX14AK53G		
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Performance Goal Text:			

Task Description:	It is well established that sleep deficiency and circadian desynchrony impair alertness and performance. It is also increasingly recognized that the response to sleep loss and circadian desynchrony is highly individualized and both challenge- and task-dependent. This high degree of inter-individual variability makes it very difficult to predict who will be at risk and when they will be at risk, and therefore a 'one-size-fits-all' approach is likely to be suboptimal. What is needed are biomarkers of individual vulnerabilities and resiliencies to sleep loss and circadian desynchronization that can be used to predict problems and intervene when necessary. In this project, we will first identify candidate biomarkers in a retrospective analysis from neurocognitive and psychological measures collected in more than 1,500 subjects that have been studied over the past 17 years in our laboratory analog and from existing extensive longitudinal data in over 50 individuals overwintering in Antarctica, which is a high fidelity space analog. These, and previously untested biomarkers, will then be validated in new limited laboratory analog studies to inform a new space analog study that will test the feasibility and utility of a set of core biomarkers to predict and assess variation in neurocognitive and psychological function associated with living in Antarctica during a 12-month mission. These innovative studies have a high potential to identify reliable biomarkers that are suitable for operational use.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of a candidate biomarker panel that delineates inter-individual differences in sleep and circadian rhythms has enormous potential for the development of individualized predictive models of vulnerabilities and resiliencies to sleep deprivation and circadian misalignment for a wide range of occupational settings, for example in shift-workers and other unusual environments (military, shipping, mining, etc).
Task Progress:	There are three main phases to our project. In Phase 1, we will identify candidate biomarkers to predict neurocognitive and psychological responses to sleep deprivation and circadian misalignment from a retrospective analysis of existing sleep, circadian, and performance data of more than 1,500 subjects that have been studied over the past 17 years. In Phase 2 of this protocol, we will identify candidate biomarkers to predict neurocognitive and psychological responses to sleep deprivation and circadian misalignment in a prospective analysis of multiple -omics data (e.g., metabolomics, lipidomics, immune response) from up to 30 subjects who will participate in a 6-day study in our laboratory analog. In Phase 3, we will assess candidate biomarkers in Antarctic expeditioners just prior to or immediately at the start of a 6-12 month overwintering period to predict individual vulnerabilities to sleep deprivation and circadian misalignment based on our findings from Phases 1 and 2. We will then collect from these individuals a number of subjective and objective measures of individual behavioral health outcomes, including sleep, circadian rhythms, urinary cortisol and melatonin levels, and a range of neurocognitive performance outcomes and alertness, to validate our predictions. In the first year of this grant, we have made significant progress in the identification of potential biomarkers from objective and subjective measures of cognitive performance and alertness. To date, we have compiled and analyzed retrospective performance and alertness data from over 160 subjects who were studied under identical laboratory conditions. From these data, we have developed a preliminary candidate biomarker panel that can predict whether an individual will be resilient or vulnerable to sleepiness-related performance impairment from information collected in the individual at baseline after a habitual (8 hour) night of sleep. This preliminary candidate biomarker panel will be used as a screening tool to select subjects for our 6-day Phase 2 study. The development of this preliminary candidate biomarker panel involved three goals. The first goal was to develop a definition of "good" vs. "poor" performance and alertness after a habitual night of sleep and less than 16 hours of wakefulness to identify those individuals that may have performance and alertness impairments unrelated to sleep loss. The second goal was to develop a definition of resilient, intermediate, and vulnerable performance impairment related to sleep loss at more than 16 hours of wakefulness. The third goal was to determine a priori which of the individuals who appear to be good performers at baseline (i.e., not susceptible to non-sleep-related performance impairment at less than 16 hours of wakefulness) were susceptible to sleep-related performance impairment at more than 16 hours of wakefulness. Our analysis revealed that among those individuals who perform well under baseline conditions, a subset remain "resilient" after more than 16 hours awake and the remaining subjects exhibit varying degrees of sleepiness-related performance impairment, some of which meet our definition for "vulnerable." The model that we have developed has been validated to predict the most "resilient" and "vulnerable" individuals with a specificity of >85%. Potential subjects that are predicted as "resilient" or "vulnerable" during our screening assessments and also meet all our other inclusion/exclusion criteria will be enrolled in our 6-day laboratory-based study in Phase 2 of this project. In this laboratory study, we will collect multiple biomarker samples during baseline and sleep deprivation to determine whether there are one or more "omics"-based measures that provide a greater sensitivity and specificity to identify individuals who are resilient and vulnerable to sleepiness-related performance impairment. We are completing the development of our screening tool and expect to start recruiting for this study by June 2015, with our first subject completing all screening procedures and entering the laboratory-based portion of the study in July 2015. We will study up to 20 subjects in Phase 2. We are also conducting the same Phase 1 analyses on our retrospective data in expeditioners overwintering in Antarctica. We will develop a similar model to predict the degree of resiliency or vulnerability in sleepiness-related performance impairment among this population of individuals, and apply this model to baseline data collected from expeditioners who will be overwintering in Antarctica from March to October of 2016. We expect to deploy this model and start baseline data collection in these individuals as early as September 2015.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Abstracts for Journals and Proceedings	St Hilaire MA, Rahman SA, Barger LK, Duffy JF, Quackenbush J, Sullivan JP, Czeisler CA, Kristal B, Lockley SW. "Development and Testing of Biomarkers to Determine Individual Astronauts' Vulnerabilities to Behavioral Health Disruptions." Presented at the 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015